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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 10: Impact of Transportation Systems-V

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Contents

- Noise
- Harmful effects of Noise
- Noise level Standards
- Factors affecting Traffic Noise
- Sources of Vehicular Noise
- Noise Measurement
- Mitigating measures for Noise Pollution
- Numerical Problems



Noise

- Noise is Unwanted sound.

Can be defined as the level of sound which exceeds the acceptable level and creates annoyance



- Urbanization and Traffic related activities are the main contributors to Noise Pollution.
- Noise from vehicles contributes to around 70% of the total Traffic Noise.
- Result of activities such as road and rail traffic, aircrafts, industrial activities and construction works.

Source: (L. R. Kadiyali, 2017)

Harmful Effects of Traffic Noise

The negative effects of Noise can be categorized into three major groups:

- Subjective effects
- Behavioral effects
- Physiological effects



Source: (L. R. Kadiyali, 2017)

Harmful Effects of Traffic Noise

Subjective effects

- Can be described by terms such as Annoyance, disturbance, or noisiness.

- Difficult to be measured precisely, as different subjects perceive a “disturbing or annoying” level of sound differently.



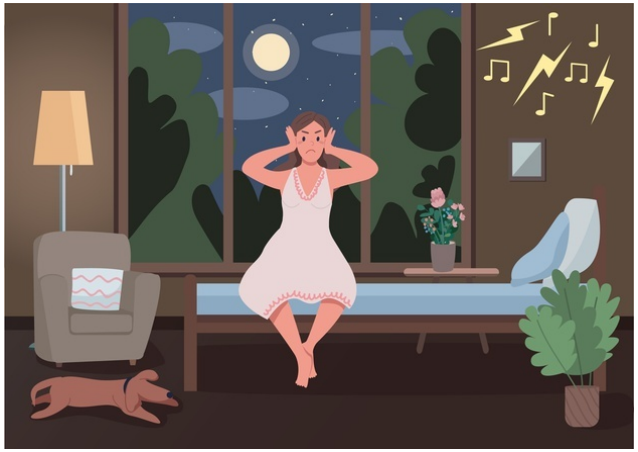
Noise level which is annoying or disturbing for some subjects might be comfortable for some other groups. (Music Vs Noise)

Source: (L. R. Kadiyali, 2017)

Harmful Effects of Traffic Noise

Behavioral effects

- Behavioral effects of Noise are related to its interference with speech, sleep or any other tasks performed.



Disturbance in sleep is common in high Noise areas, such as dense traffic areas or near railway tracks or airports.



Source: (L. R. Kadiyali, 2017)

Harmful Effects of Traffic Noise

Physiological effects

- Physiological effects of Noise are those that can cause a sudden shock or frightening effect.
- May cause harmful effects on various parts of the body, e.g., headache.
- Exposure to extremely high level noise for longer duration can even cause deafness.



Source: (L. R. Kadiyali, 2017)

Harmful Effects of Noise on Human Beings

- Interferes with speech. (Ex. A person may not be able to follow or understand what the other person might be saying)
- Emotional and Behavioral Stress. (Ex. A person might feel disturbed or stressed due to annoying sounds)
- Permanent hearing loss. (Ex. Continuous and prolonged exposure to large noise can severely impact human eardrum)
- Potential to cause diseases. (Ex. Headache, blood pressure, heart failure etc.
- Potential to cause disturbing health effects. (Ex. Increased heart beat, constriction of blood vessels, dilation of pupil etc.)
- Problematic for patients who needs rest.
- Potential to damage liver, brain and heart.



Source: (Prof. Anil Kumar, IIT Roorkee)

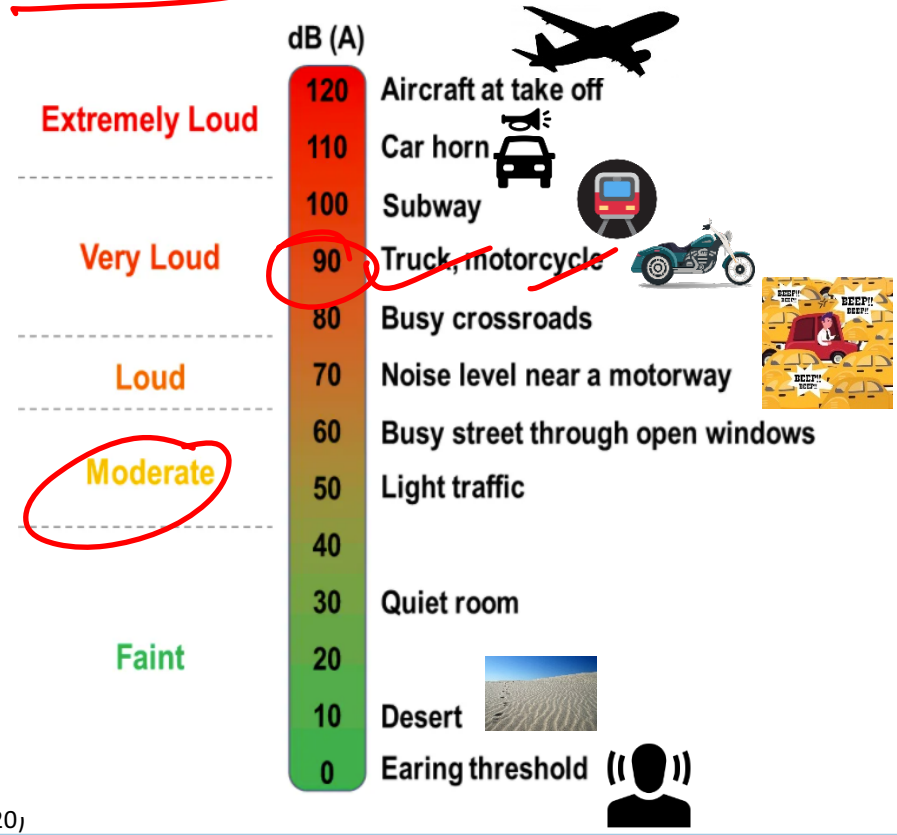
Noise Level Standards

S. No	Area description	Noise level in dB(A)			
		FHWA	AASHTO	CPCB, India	
				Day time (6.00 AM – 9.00 PM)	Night time (9.00 PM – 6.00 AM)
1	Sensitive areas such as Parks, schools, hospitals, and silent zones	60	55-60	50	40
2	Residential areas	70 (Interior max. 55)	70 Exterior 55 Interior	55	45
3	Mixed areas	70	70	---	---
4	Commercial areas	75	75	65	55
5	Industrial areas	75	75	75	70

FHWA: Federal Highway Administration (USA)
AASHTO: American Association of State Highway and Transportation Officials (USA)
CPCB: Central Pollution Control Board (India)

Source: (Prof. Anil Kumar, IIT Roorkee)

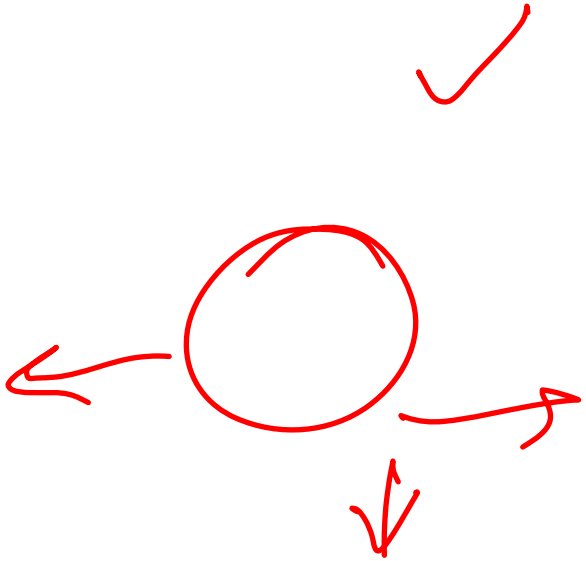
Typical Noise Levels



Noise levels from different sources

Source: (Jean-Paul Rodrigue, 2020)

Factors affecting Traffic Noise





Source: (Prof. Anil Kumar, IIT Roorkee)

Factors affecting Traffic Noise

Traffic Factors



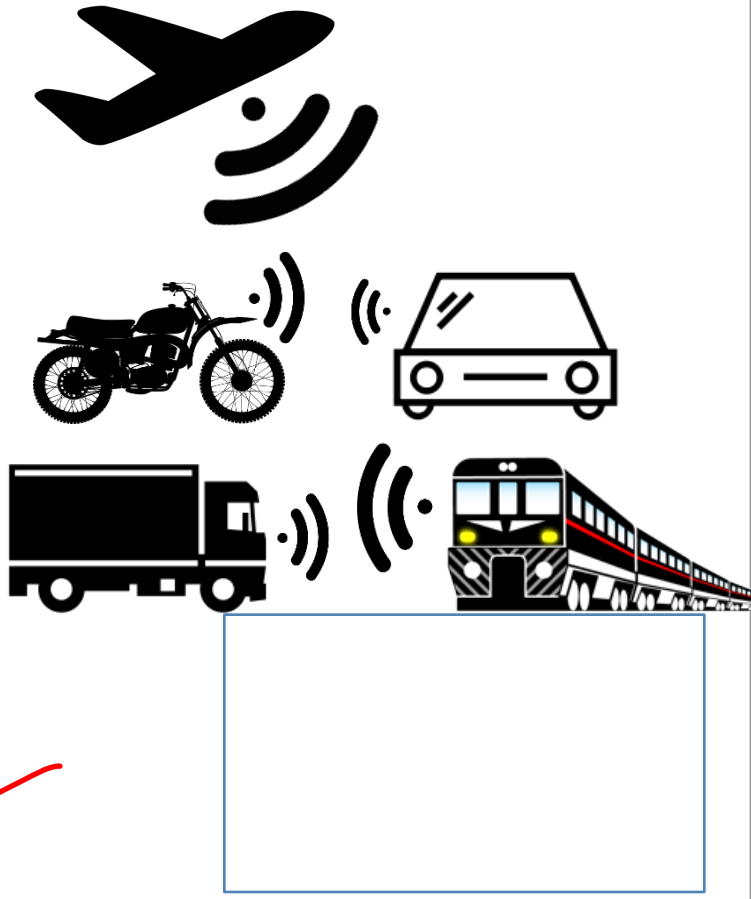


Source: (Prof. Anil Kumar, IIT Roorkee)

Sources of Vehicular Noise

Noise from Traffic vehicles: Some facts

- Motor vehicles are more noisier than passenger cars
- Diesel engines are more noisier than petrol engines
- Commercial trucks and loaded vehicles are more noisier than other road traffic modes.
- Noise is directly proportional to the age of the vehicle (Older vehicles generates more noise).



Source: (L. R. Kadiyali, 2017)

Factors affecting Traffic Noise

Road Factors











Source: (Prof. Anil Kumar, IIT Roorkee)

Factors affecting Traffic Noise

Vehicle Factors









Source: (Prof. Anil Kumar, IIT Roorkee)

Factors affecting Traffic Noise

Human Factors





Source: (Prof. Anil Kumar, IIT Roorkee)

Factors affecting Traffic Noise: Human Factors



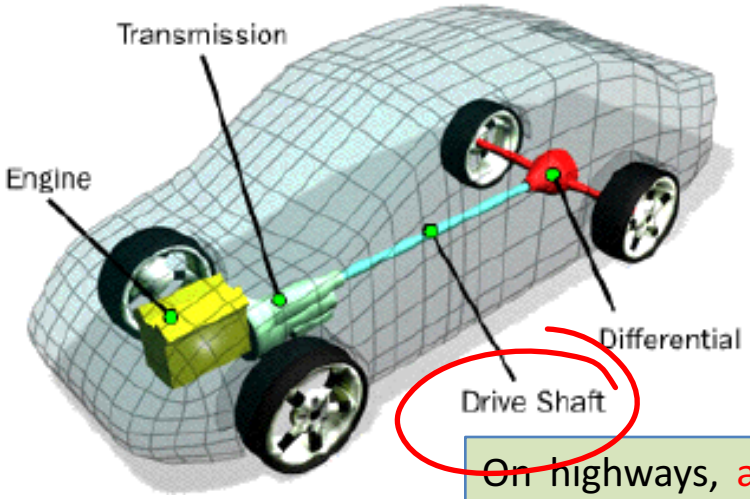
Source: (https://www.intechopen.com/books/noise-and-environment/traffic-noise)

Irresponsible and Inappropriate behavior

- 90% of accidents are linked to Human error

Sources of Vehicular Noise

Noise from Traffic vehicles on Road



Two major sources from vehicles:

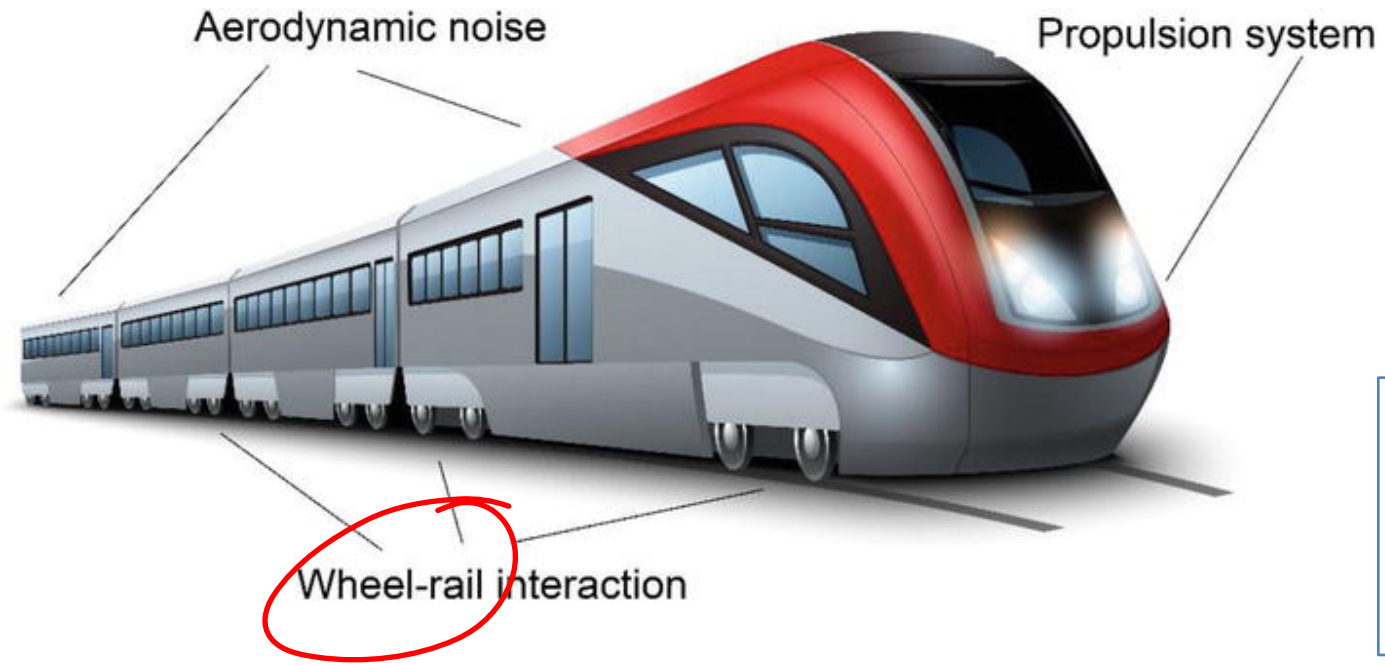
- Power plant and Transmission Noise sources: Engine, exhaust, intake, cooling system etc.
- Running gear Noise sources: Tyre-road interaction, differential and propeller shaft etc.

On highways, at lower speeds, Engine-exhaust noise is dominant and at higher speeds, noise from road-tyre interaction becomes dominant sources

Source: (Prof. Anil Kumar, IIT Roorkee)

Sources of Vehicular Noise

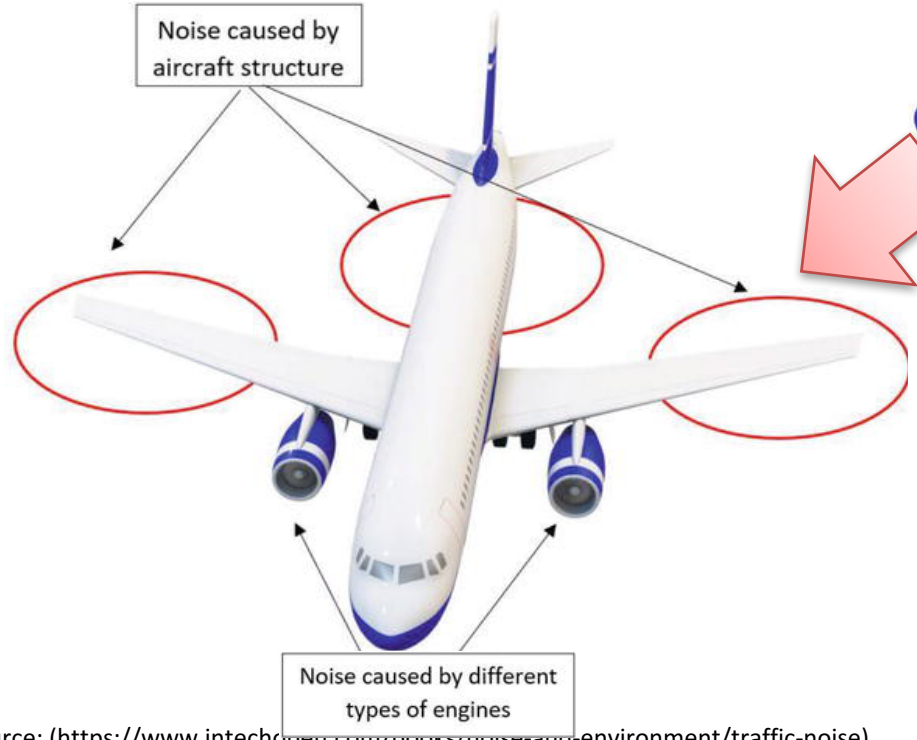
Noise from Traffic vehicles on Rail

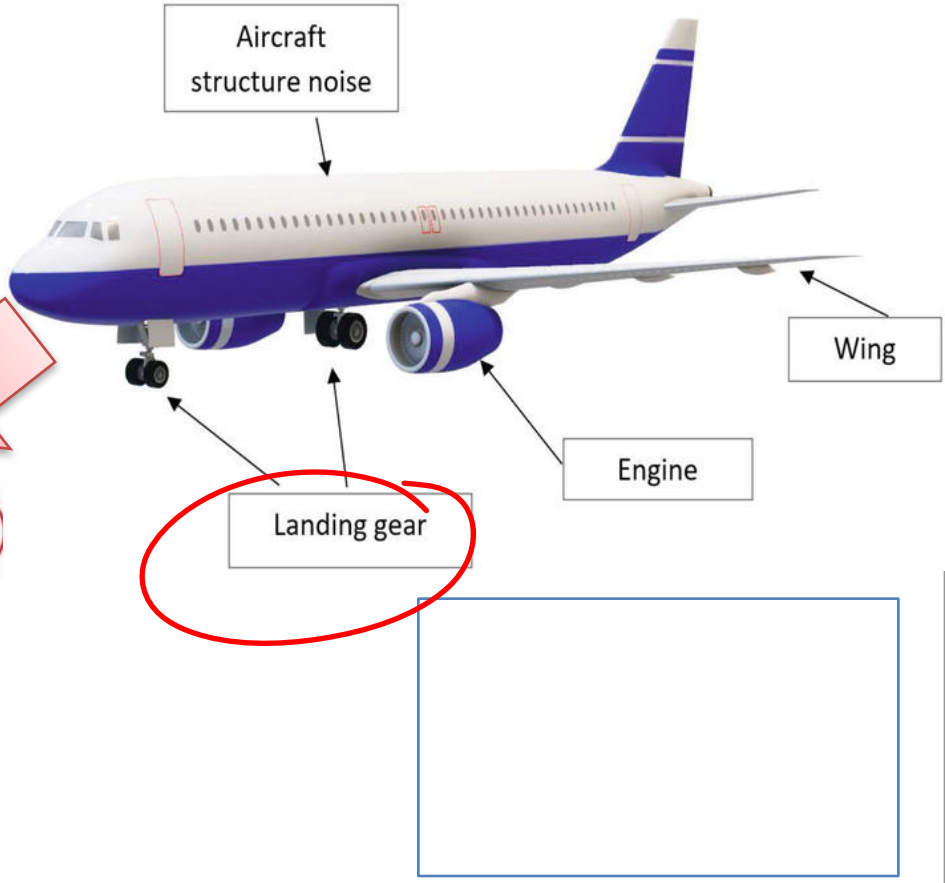


Source: (https://www.intechopen.com/books/noise-and-environment/traffic-noise)

Sources of Vehicular Noise

Noise from Traffic vehicles on Air





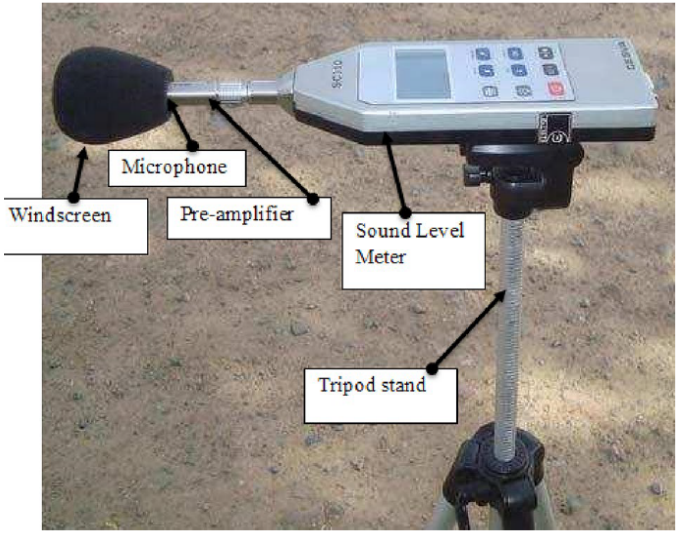
Source: (https://www.intechopen.com/books/noise-and-environment/traffic-noise)

Noise Measurement

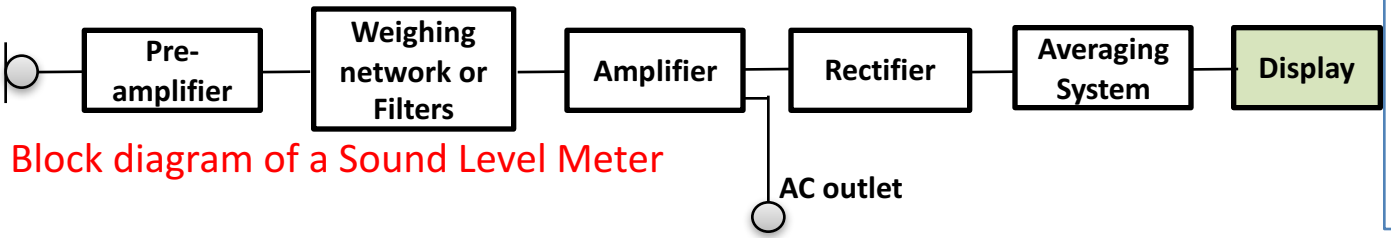
Sound Level Meter



- The Noise waves from the sources reaches the microphone.
- The electrical signal from the transducer is fed to the pre-amplifier or filters, if necessary.
- Further amplification prepares the signal for rectification and direct reading on the meter.



A Sound Level Meter Source: (Diljeet Singh et. al, 2020)



Block diagram of a Sound Level Meter

Source: (https://www.who.int/occupational_health/publications/noise6.pdf)

Noise Measurement

Acoustic Arrays



- Number of microphones are arranged in various patters.
- Omnidirectional
- Can receive sound from a wide range of area.

Source: (Prof. Anil Kumar, IIT Roorkee)

Noise Measurement

DeciBel (dB)

- Sound pressure level (SPL) or acoustic pressure level is a logarithmic measure of the effective pressure of a sound relative to a reference value.

- SPL, denoted by L_p and is measured in decibels (dB).

$$L_p = \ln\left(\frac{p}{p_0}\right) N_p = 2\log_{10}\left(\frac{p}{p_0}\right) B = 20\log_{10}\left(\frac{p}{p_0}\right) dB$$

Where,
P = root mean square sound pressure
P_0 = reference sound pressure
1B = 10 dB

Source: (Prof. Anil Kumar, IIT Roorkee)

The commonly used reference sound pressure in air is P_0 = 20 µPa, which is considered as the threshold of human hearing (roughly the sound of a mosquito flying 3m away)

Noise Measurement

Equivalent continuous Noise level (Leq)

Leq is the preferred method to describe sound levels that vary over time, resulting in a single decibel value, which takes into account of the total sound energy over a period of time of interest.

- Noise levels fluctuates over a wide range with time.
  - Ex. Sound levels fluctuates over a period of 24 hours, going as low as 30 dB(A) at night time, rising to 70 dB(A) with passing of occasional vehicles and to higher levels in day time.

How to measure an overall value for all these noise levels ?

Answer to the question is the "Equivalent continuous Noise level meter (Leq)"

Source: (Prof. Anil Kumar, IIT Roorkee)

Noise Measurement

Equivalent continuous Noise level meter (Leq)

- Equivalent continuous Noise level meter reads all the fluctuations over a period of time, stores them in its memory and at the end of measurement, calculates an "Average energy" or "Leq value".

> The "average" values are not arithmetic average, as we are measuring sound, in decibels, which are logarithmic values.

- The decibel (dB) values are converted to Sound pressure levels (SPL), adds these values and then divide by the number of samples and is finally converted back the equivalent level to decibels (dB).

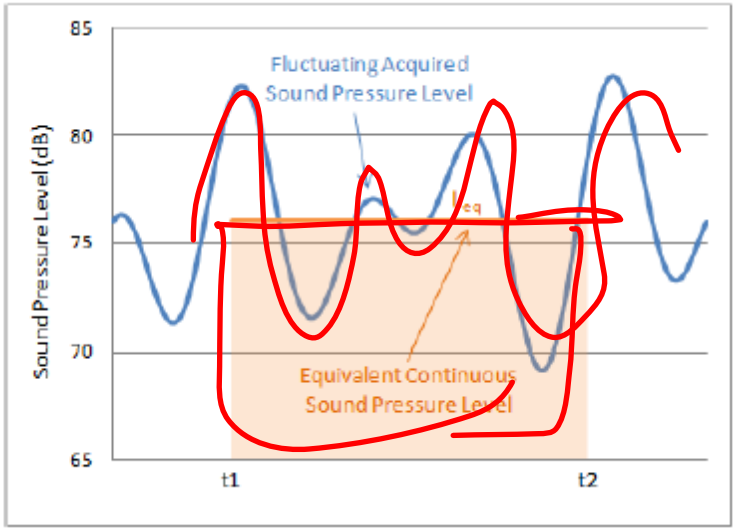
Source: (Prof. Anil Kumar, IIT Roorkee)

Noise Measurement

Equivalent continuous Noise level (Leq) Calculation

- Equivalent continuous Noise level

$$L_{eq} = 10 \log \left[\frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \frac{{p_A}^2}{{p_0}^2} dt \right]$$



where,

L_{eq} = equivalent continuous sound pressure level in dB

p_0 = reference pressure level (typically 20 μPa)

p_A = acquired sound pressure

t_1 = start time for measurement

t_2 = end time for measuremen

The equation divides the integrated, normalized sound pressure by the duration of interest of the signal.

The result is expressed in decibels:

Source: (Prof. Anil Kumar, IIT Roorkee)

Noise Measurement

Energy Average (Effective or rms sound level), Lrms Sound level

Equivalent Continuous sound level (Leq) the steady sound level of a noise energy-averaged over time

Energy-average

The energy-average (effective or rms value) of sound level is:

$$L_{rms} = 10 \log_{10} \{1/n [10^{L1/10} + 10^{L2/10} + 10^{L3/10} + \ldots + 10^{Ln/10}]\}$$

Example:

The averages of three sound levels 88, 93 ad 97 dBA are:

Arithematic Mean: Lmean = {88 + 94 + 97} / 3 = 93 dBA

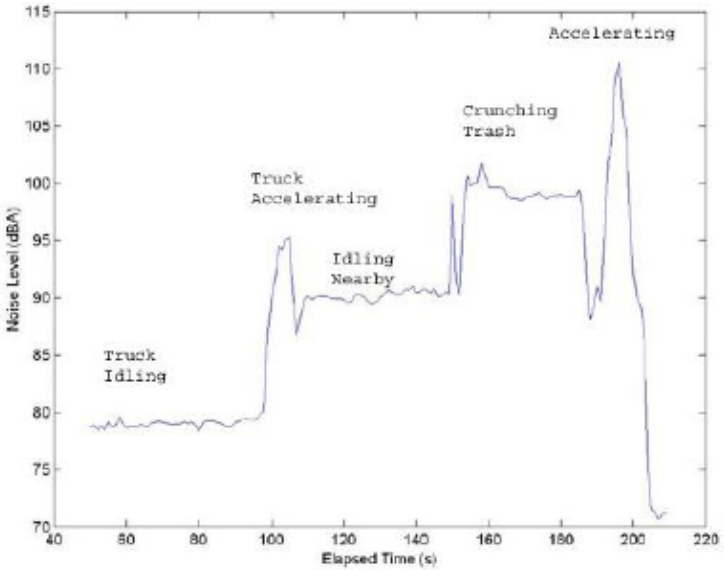
Energy-average: Leq = $10 \log_{10} \{[10^{8.8} + 10^{9.4} + 10^{9.7}]^{1/3}\}$ = 94.3 dBA

Source: (Prof. Anil Kumar, IIT Roorkee)

Noise Measurement

A-Weighting, dB(A)

A-weighting cuts off the lower and higher frequencies that an average person cannot hear



- Most common weighting used in Noise measurements.
- Unit of measured A-weighted values are expressed as dBA or dB(A).

Sound Level Meter reading vs time for a passing truck dBA or dB(A)

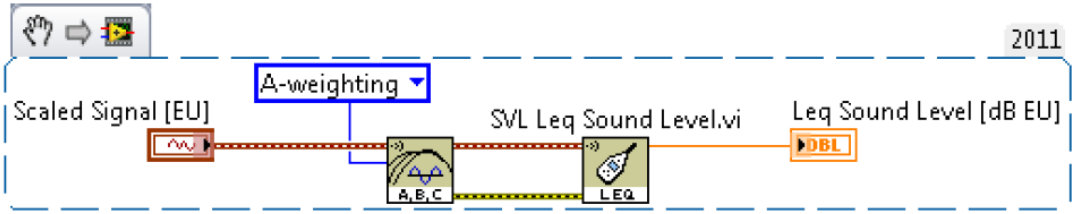
Source: (Prof. Anil Kumar, IIT Roorkee)

Noise Measurement

A-Weighting, dB(A)

- Equivalent continuous A-weighted sound pressure level is commonly used to measure noise levels in loud environments.
- Imitates the hearing profile of the Human ear.

The human ear responds more to frequencies between 500 Hz and 8 kHz and is less sensitive to very low-pitch or high-pitch noises.



- The *frequency weightings* used in sound level meters are often related to the response of the human ear, to ensure that the meter is measuring pretty much what you actually hear.

Source: (Prof. Anil Kumar, IIT Roorkee)

Noise Measurement

The n-Percent Exceeded Level, L_n

- The n-percent exceeded level, L_n, is the sound pressure level exceeded for n% of the time.

- The fluctuating sound pressure levels are higher than the L_n level.

L_{10}: > L_{50} > L_{90}



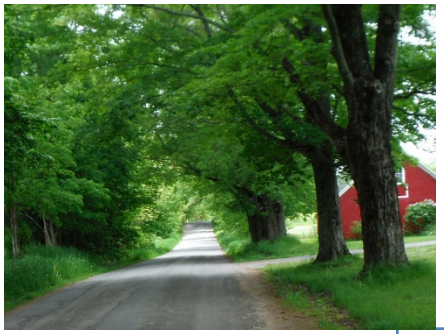
- L_{10}: The sound level exceeded for 10% of the time. (Rest of the time, the sound or noise has a pressure level at or below , L_{10}). Due to sporadic or intermittent events.
- L_{50}: The sound level exceeded for 50% of the time. Represents median or fluctuating noise.
- L_{90}: The sound level exceeded for 90% of the time. Represents background or ambient level of noise.

Source: (Prof. Anil Kumar, IIT Roorkee)

Mitigating measures for Noise Pollution

- Green plantations: Planting bushes and trees around noise generating sources
- Regular servicing: Regular servicing and tuning of automobiles
- Green Buildings: Use of suitable noise absorbing materials for walls, windows and ceilings
- Wearables: Equipments such as ear plugs and earmuffs for hearing protection









Source: (Prof. Anil Kumar, IIT Roorkee)

Mitigating measures for Noise Pollution

- Lubrication and regular servicing of machinery, similar to automobiles
- Policy regulations: Policy measures to restrict use of loudspeakers in crowded and public areas
- Soundproof materials: Use of soundproof doors and windows to block unwanted outside noise.
- Land use: Factories and industries should be located far from residential areas

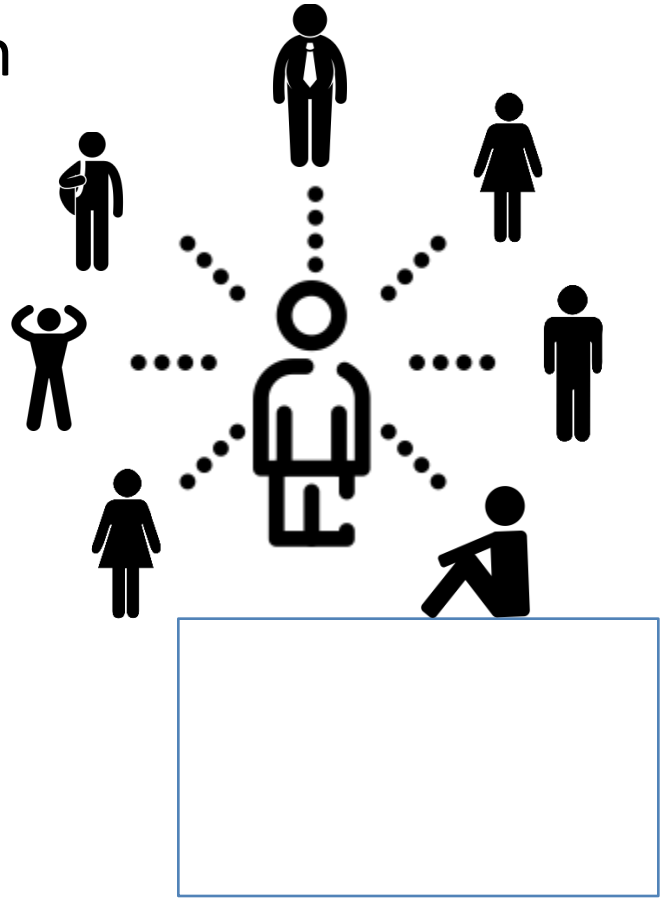


Source: (Prof. Anil Kumar, IIT Roorkee)

Mitigating measures for Noise Pollution

- Community development programs: Community development or Urban management should be done with long term planning to reduce noise pollution

- Social awareness programs: To educate and to create awareness among public for long term noise reduction

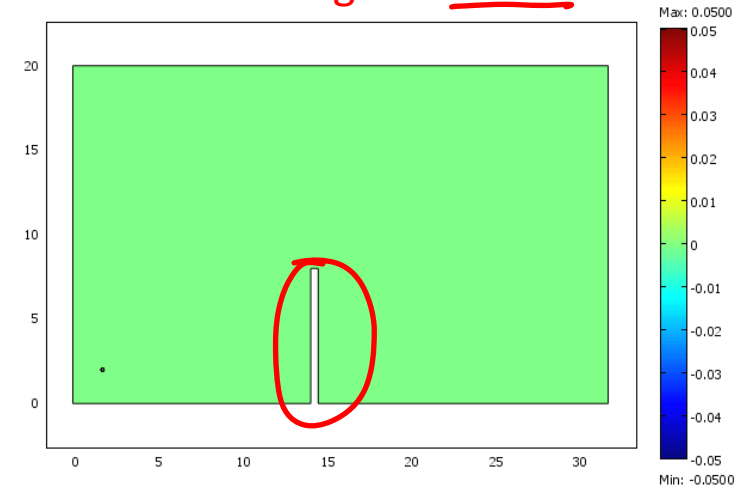


Source: (Prof. Anil Kumar, IIT Roorkee)

Noise Barriers

- Noise barriers reduce the sound level by blocking the straight-line path from the source to the receiver.
- Noise level can be reduced by about 5dB.
- The noise levels can further be reduced by 3 dB by forcing the sound to travel a longer distance by making the barrier much higher.
- The combined effect (noise reduction by 8dB) is equivalent to reducing the traffic by a factor of 6.

Barrier height h = λ



Diffraction over a Noise barrier varies with the ratio of barrier height (h) to wavelength (λ)

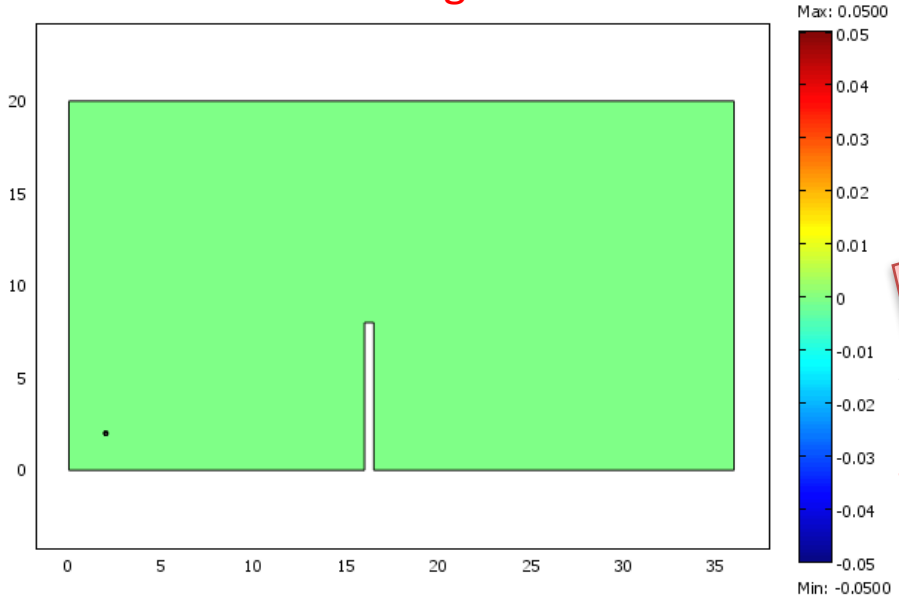
Source: (https://mypages.iit.edu/~muehleisen/acs_demos/diffraction/diffraction.html)

Source: (Prof. E. Rajasekar, IIT Roorkee)

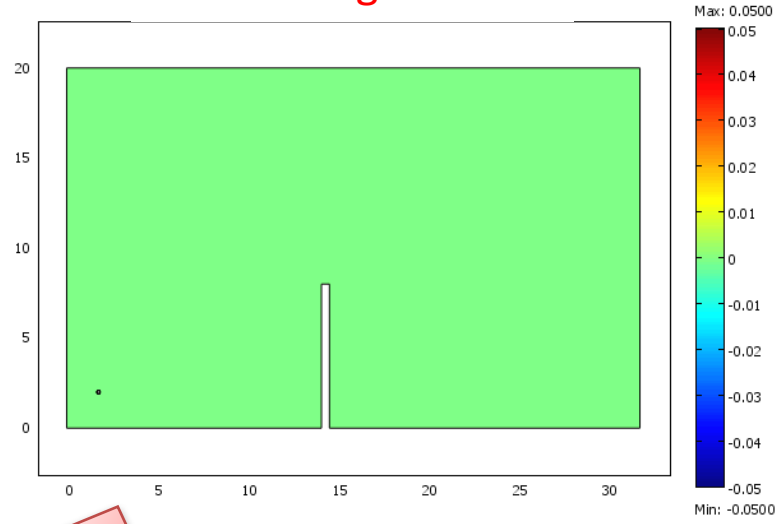
Noise Barrier Effectiveness

Diffraction of Noise with respect to the ratio of barrier height (h) to wavelength (λ)

Barrier height h = 2λ



Barrier height h = 4λ



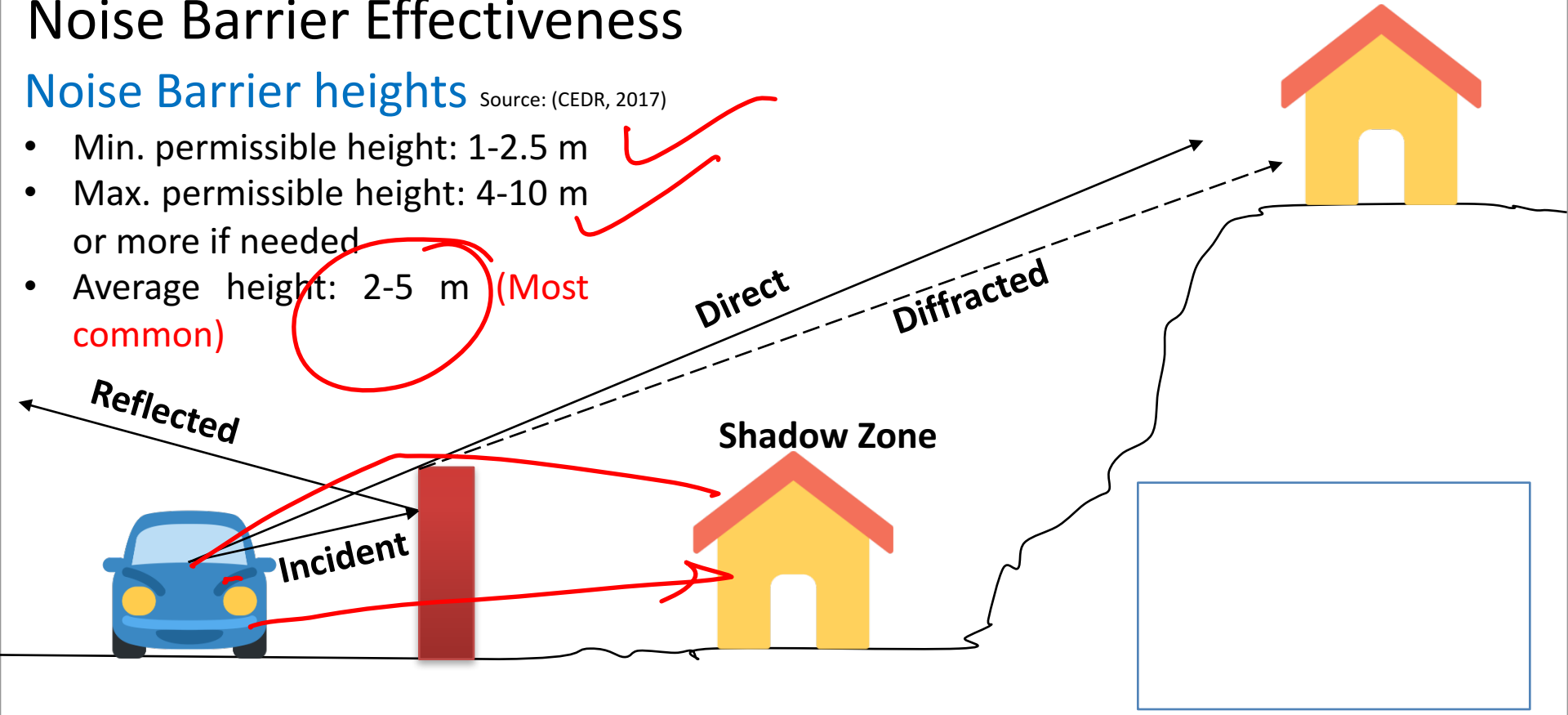
As height (h) of Noise barriers increases, the efficiency of the barrier also increases

Source: (https://mypages.iit.edu/~muehleisen/acs_demos/diffraction/diffraction.html)

Noise Barrier Effectiveness

Noise Barrier heights Source: (CEDR, 2017)

- Min. permissible height: 1-2.5 m
- Max. permissible height: 4-10 m or more if needed
- Average height: 2-5 m (Most common)



Noise Barrier: Examples

Timber acoustic fencing



Good quality acoustic fencing will stay imperforate over time

Hedging, foliage



Dense, deep foliage will provide some acoustic benefit but line hedges are ineffective

Source: (https://www.cassallen.co.uk/guide-noise-barriers)

Noise Barrier: Examples

Brick walls



Good, but expensive compared to timber fencing.

Green screens



Gaps render green screens acoustically transparent

Source: (https://www.cassallen.co.uk/guide-noise-barriers)

Noise Barrier: Examples

Transparent acoustic screens (Glass or acrylic)



Good where visual amenity is important

Hit & miss fencing



Ineffective due to gaps

Source: (https://www.cassallen.co.uk/guide-noise-barriers)

Noise Barrier: Examples

Concrete



Low maintenance, but expensive.

Light weight metal fencing



Ensure minimum mass of $10kg/m^2$.

Source: (https://www.cassallen.co.uk/guide-noise-barriers)

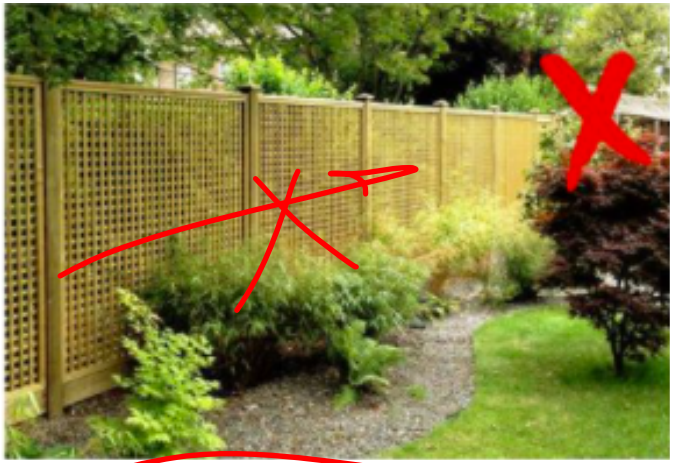
Noise Barrier: Examples

Bunding



Suited to larger projects where spoil can be used to form bunds.

Trellising



Ineffective due to gaps

Source: (https://www.cassallen.co.uk/guide-noise-barriers)

Noise Barrier: Examples

Gabion walls



Important that the wall is thick and the aggregate is small to be acoustically effective.

Loosely lapped fencing



Some screening will be provided, but gaps will affect the performance.

Gaps tend to increase over time.

Source: (https://www.cassallen.co.uk/guide-noise-barriers)

Noise Barrier: Examples

Fully bespoke/engineered solutions



Suited to larger infrastructure projects- Very expensive

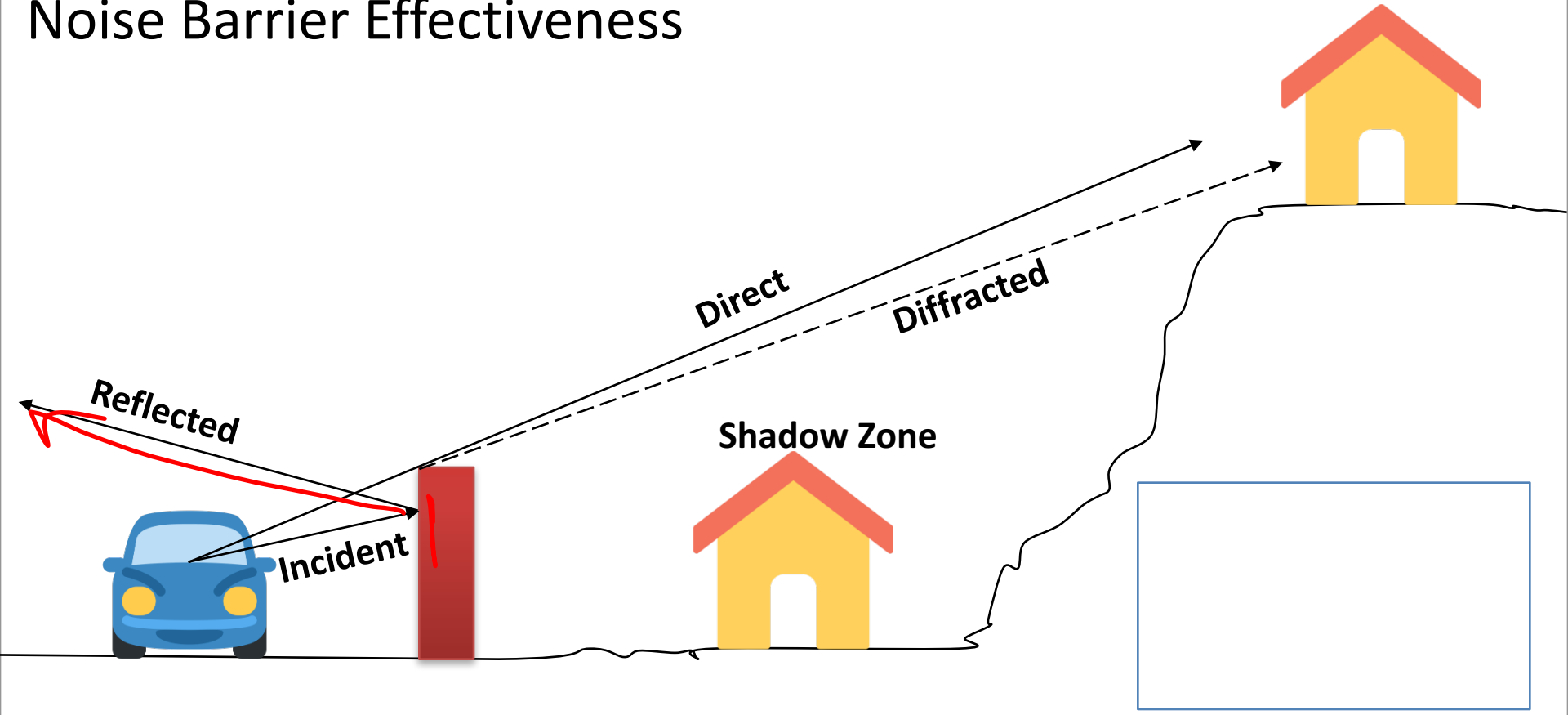
Slatted fencing



Ineffective due to gaps

Source: (https://www.cassallen.co.uk/guide-noise-barriers)

Noise Barrier Effectiveness



Noise Barrier Losses

Transmission Loss

- The transmission loss is the sound level that gets reduced, when it is forced to travel through the barrier.

- A 30 dB transmission loss means practically all (99.9%) of the sound is being blocked.

Insertion Loss

- Insertion level is the difference in sound levels, if there were no barrier present and the resultant sound level after a barrier is installed between the noise source and the receiver.

Source: (Prof. E. Rajasekar, IIT Roorkee)

Insertion Loss and Fresnel Number

- Insertion loss (IL) is estimated by using the model proposed by Kurze and Anderson.

$$\text{Insertion Loss, IL} = 5\ \text{dB} + 20 \log \left[\frac{\sqrt{2\pi N}}{tanh\sqrt{2\pi N}}\right] dB \quad for\ (N\ upto\ 12.5)$$

$$\text{Insertion Loss, IL} = 20\ \text{dB} \quad for\ N > 12.5$$

- N is the Fresnel Number, a non-dimensional measure of how much farther the sound must travel to reach the receiver, as a result of the barrier.
- N is calculated using the equation, $N = \frac{(a+b-l)f}{C_0}$

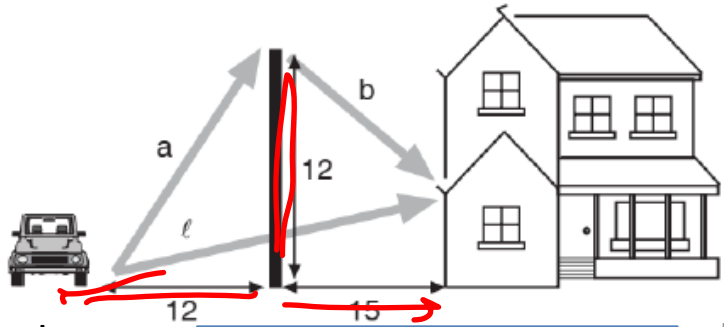
where,
l = original length of direct path from source to receiver
a and b are the lengths of two straight line segments, comprising the path as modified by the barrier
F is the sound frequency in Hertz (Hz); C_0 = speed of sound in air (approx. 1125 ft/sec or 343 m/s)

Source: (Prof. E. Rajasekar, IIT Roorkee)

Numerical problem

Question:

A 12 ft tall Noise barrier is installed 12 ft from the vehicle. A house is 15 ft beyond the barrier and has a window at a height of 4 ft. Calculate the Insertion loss ?



Solution:

The length of the original path of sound (l)

$$l = \sqrt{27^2 + 4^2} = 27.3 \text{ feet}$$

The length of the segments (a & b)

$$a = \sqrt{12^2 + 12^2} = 17 \text{ feet}$$

$$b = \sqrt{15^2 + 8^2} = 17 \text{ feet}$$

Source: (Prof. E. Rajasekar, IIT Roorkee)

Numerical problem

Solution (cont'd):

Fresnel number, $N = \frac{(a+b-l)f}{C_0}$

Hence,

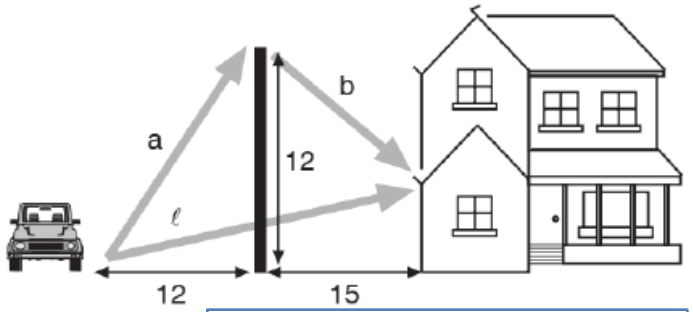
$(a + b - l) = 34\text{-}27.3 = 6.7$ feet

At f = 100 Hz, the Fresnel number, N is;

$$N = \frac{6.7 \; x \; 100}{1100} = 0.61$$

The Insertion loss (IL) can be calculated by:

$$IL = 5 + 20 \log \left[\frac{\sqrt{2\pi \; x \; o.61}}{tanh\sqrt{2\pi \; x \; 0.61}}\right] \approx 10 \text{ dB}$$



Source: (Prof. E. Rajasekar, IIT Roorkee)

Conclusion

- Loud and regular exposure to Noise can have significant impacts on living beings including humans from slight disturbances to even death.

- Measures to reduce Noise emissions should be introduced from the planning stage of a project.

- Use of Noise absorbing materials such as trees and shrubs on road-sides, installation of noise barriers etc. can reduce the impacts of Noise.

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Thank You